INCH-POUND

MIL-C-55302/58H
23 September 1992
SUPERSEDING
MIL-C-55302/58G
26 August 1988

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, RECEPTACLE, SOCKET CONTACTS, DECADE INCREMENTS 10 THRU 70 CONTACT POSITIONS AND INTERMEDIATE POSITIONS OF 14, 24, 44, 54 AND 26, 36, 56, 66 FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for aquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55302.

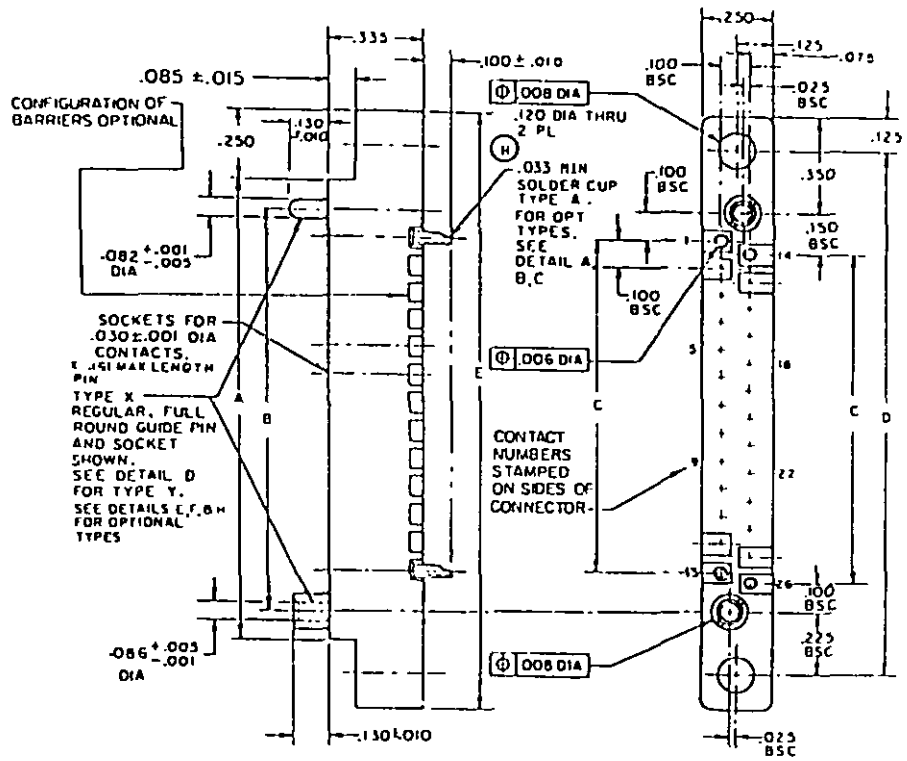


FIGURE 1. Connector, receptacle .100 (2.54 mm) spacing.

(H) denotes changes

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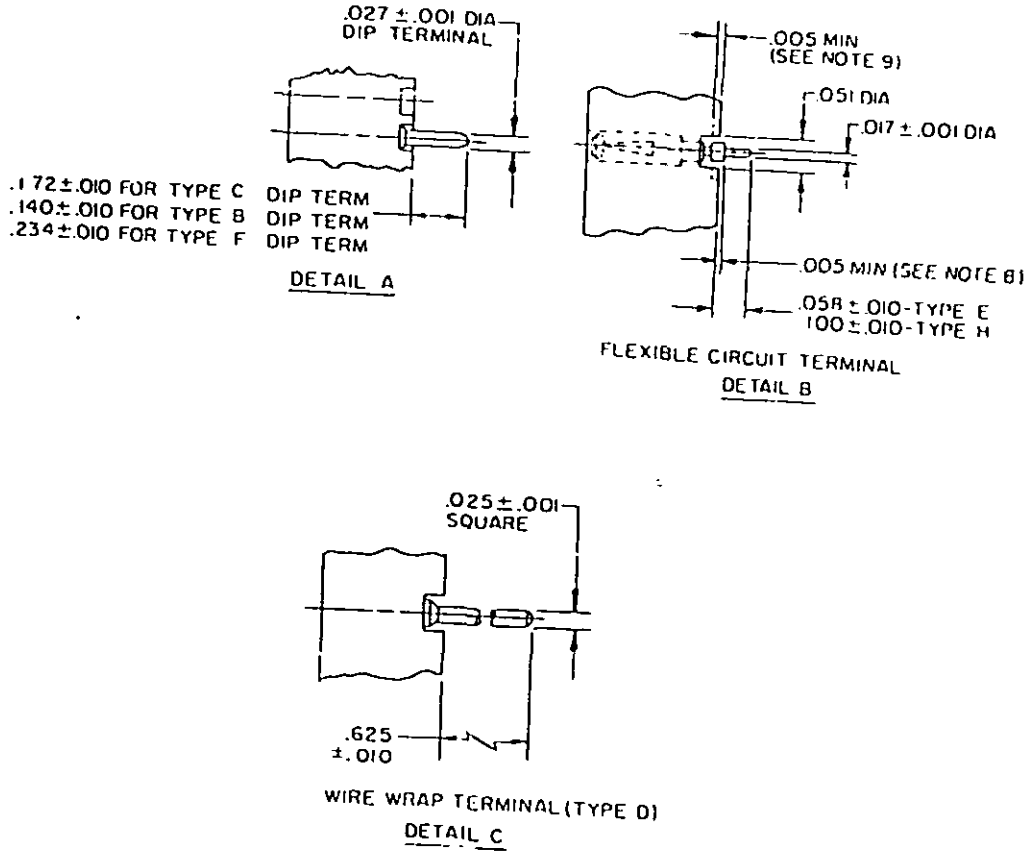
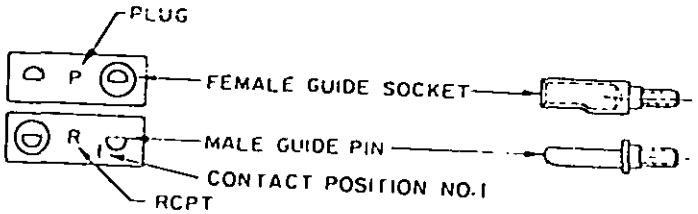


FIGURE 1. Connector, receptacle .100 (2.54 mm) spacing - Continued.

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CONNECTOR POLARIZATION WITH D SHAPED GUIDE PINS AND SOCKETS. (TYPE Y)

DETAIL D

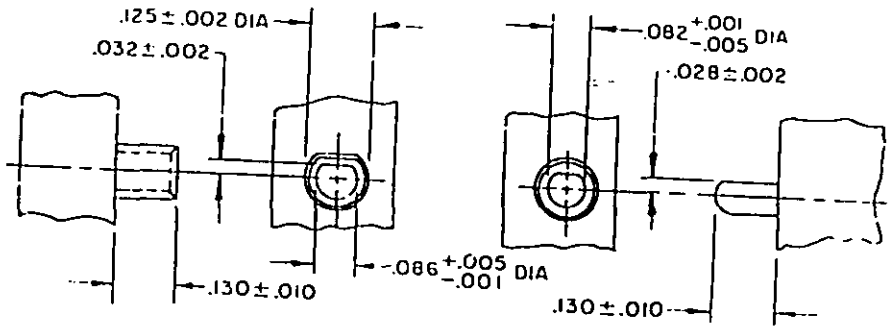
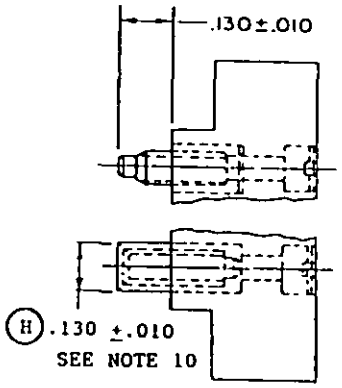
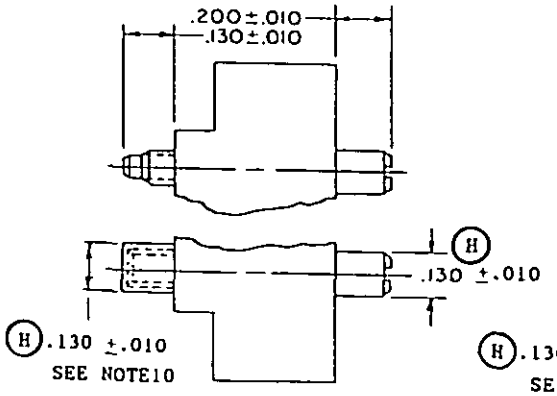


FIGURE 1. Connector, receptacle .100 (2.54 mm) spacing - Continued.

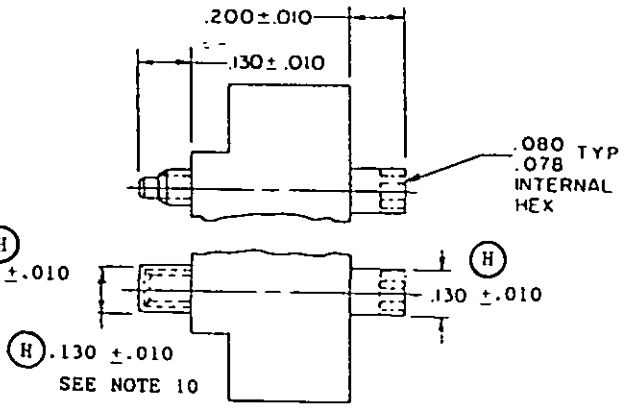
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DETAIL E



DETAIL F



DETAIL H

FIGURE 1. Connector, receptacle .100 (2.54 mm) spacing - Continued.

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Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.017	0.43
.025	0.64
.027	0.69
.028	0.71
.030	0.76
.035	0.89
.050	1.27
.051	1.30
.058	1.47
.075	1.91
.078	1.98
.082	2.08
.086	2.18
.094	2.39
.100	2.54
.108	2.74
.120	3.05
.125	3.18
.130	3.30
.140	3.56
.150	3.81
.161	4.09
.172	4.37
.200	5.08
.225	5.72
.250	6.35
.335	8.51
.350	8.89
.625	15.88

Number of contacts	Dimensions				
	A ±.010	BSC B	REF C	BSC D	E ±.010
10	.850 (21.59)	.650 (16.51)	.400 (10.16)	1.100 (27.94)	1.350 (34.29)
14	1.050 (26.67)	.850 (21.59)	.600 (15.24)	1.300 (33.02)	1.550 (39.37)
20	1.350 (34.29)	1.150 (29.21)	.900 (22.86)	1.600 (40.64)	1.850 (46.99)
24	1.550 (39.37)	1.350 (34.29)	1.100 (27.94)	1.800 (45.72)	2.050 (52.07)
26	1.650 (41.91)	1.450 (36.83)	1.200 (30.48)	1.900 (48.26)	2.150 (54.61)
30	1.850 (46.99)	1.650 (41.91)	1.400 (35.56)	2.100 (53.34)	2.350 (59.69)
36	2.150 (54.61)	1.950 (49.53)	1.700 (43.18)	2.400 (60.96)	2.650 (67.31)
40	2.350 (59.69)	2.150 (54.61)	1.900 (48.26)	2.600 (66.04)	2.850 (72.39)
44	2.550 (64.77)	2.350 (2.100)	2.100 (53.34)	2.800 (71.12)	3.050 (77.47)
50	2.850 (72.39)	2.650 (67.31)	2.400 (60.96)	3.100 (78.74)	3.350 (85.09)
54	3.050 (77.47)	2.850 (72.39)	2.600 (66.04)	3.300 (83.82)	3.550 (90.17)
56	3.150 (80.01)	2.950 (74.93)	2.700 (68.58)	3.400 (86.36)	3.650 (92.71)
60	3.350 (85.09)	3.150 (80.01)	2.900 (73.66)	3.600 (91.44)	3.850 (97.79)
66	3.650 (92.71)	3.450 (87.63)	3.200 (81.28)	3.900 (99.06)	4.150 (105.41)
70	3.850 (97.79)	3.650 (92.71)	3.400 (86.36)	4.100 (104.14)	4.350 (110.49)

NOTES:
1. Dimensions are inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. These connectors mate with connectors specified in MIL-C-55302/57, /61, and /63 using appropriate hardware.
5. Numbers indicating the first and last position in each row and every fourth contact position in between shall be marked on the side(s) of the connector. As an option to the above, numbers indicating every fourth cavity may be stamped on the side of the connector, with the exception that the number one contact shall be marked.
6. Termination layout on .025 (0.64 mm) modular grid.
7. Metric equivalents are in parentheses.
8. Conformal coating clearances shall be .005 (0.13 mm) minimum (not required for optional design).
9. Optional design conformal coating clearance.
(H) 10. Hole depth to female guide hardware is .282 min (7.16 mm). Full thread depth to female threaded hardware is .240 min (6.1 mm).

FIGURE 1. Connector, receptacle .100 (2.54 mm) spacing - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Material: Guide pins, type X guide sockets, jackscrews, and jacksockets shall be made of corrosion resisting stainless steel in accordance with ASTM-A581 or ASTM-A582 passivated in accordance with QQ-P-35 or shall be corrosion resistant copper nickel alloy (61 ±2% copper, 25 ±2% nickel, 2.5% max. other, and the balance zinc) and shall have a brinell hardness of 145 to 175. Type Y guide sockets shall be beryllium copper in accordance with QQ-C-550, nickel plated in accordance with QQ-N-290, class 2, 50 to 150 microinches.

Plating: The contact plating shall be in accordance with MIL-C-55302 or gold in accordance with MIL-G-45204, type II, grade C, class 1, over nickel plating in accordance with QQ-N-290, class 2, 50 to 150 microinches in the engaging area and gold in accordance with MIL-G-45204, type II, grade C, class 00, or type III, grade A, class 00, over nickel plating in accordance with QQ-N-290, class 02, 50 to 150 microinches in the termination area.

Contact identification: See figure 1.

Contact engagement and separation forces (number 22 contacts):

Minimum separation: 0.5 ounce.
Maximum engagement: 12.0 ounces for standard force and 4.0 ounces for low force.
Note: Low force contacts meet the requirements of standard force contacts.

Mating and unmating:

Standard insertion force contacts: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.56 and the withdrawal force, in pounds, shall be a minimum of 0.08 times the number of contacts and shall not exceed the measured insertion force.

Low insertion force contacts: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.25 and the withdrawal force, in pounds, shall be a minimum of 0.04 times the number of contacts and shall not exceed the maximum insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Dielectric withstanding voltage:

Sea level: 1,000 volts rms.
High altitude: 300 volts rms.

Current rating: 5.0 amperes maximum per contact.

Polarization: Type X, regular, guide pin and socket, full round (see figure 1). Type Y, adjustable "D" shaped guide pin and/or socket (see MIL-C-55302/57, detail B). All connectors are supplied in the -1 polarized position. For other than OEM use, connectors may be repolarized by using M55302/57-01 spanner wrench and M55302/57-02 guide pin and guide socket kit. When guides are removed and reinstalled in another position apply Loctite 83-CV or equivalent to the threads. Ordering connectors in other than the -1 polarized position may be done by the OEM's by placing the desired polarized position in brackets three spaces after the part number on the purchase order. Example M55302/58-A10Y -32 . Polarization pins and sockets shall be adjusted to any desired configuration by spanner wrench (M55302/57-01). Ordering connectors in the -1 polarized position without Loctite 83-CV or equivalent applied may be done by placing a "-0" in brackets three spaces after the part number on the purchase order. Guide sets, as installed, shall withstand 7 inch ounces backoff torque applied through the guide set mounting nut.

M55302/57-02 kit: Consists of one type Y "D" shaped guide socket and one type Y "D" shaped guide pin.

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Example of part number:

M55302/58	L	A	10	X
Basic number of specification sheet	Type of contacts	Type of terminals (see figure 1)	Number of contacts (see figure 1)	Type of mounting hardware (see details B, C, D, and E of figure 1)

L = Low insertion force contacts
- = Standard insertion force contacts

A = solder cup
B = dip terminal .140 long
C = dip terminal .172 long
D = wire wrap
E = .058 flexible circuit terminal
F = dip terminal .234 long
H = .100 flexible circuit terminal

Type X = full round
Type Y = "D" shaped
Type F = fixed jackscrew
Type S = .200 slotted turning jackscrew
Type H = .200 hex turning jackscrew

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
(H) NASA - NA

Review activities:
Army - AR, MI
Navy - AS, TD
Air Force - 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC, OS
Air Force - 11, 19, 80

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3860-4)

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